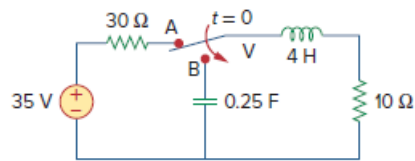


Problem

The switch in Fig. 16.42 moves from position A to position B at $t = 0$ (please note that the switch must connect to point B before it breaks the connection at A, a make before break switch). Find $v(t)$ for $t > 0$.

Figure 16.42:



Step-by-step solution

Step 1 of 8

Refer to Figure 16.42 in the text book.

Draw the circuit diagram at $t < 0$.

At $t < 0$, inductor acts as short circuit and capacitor acts as open circuit.

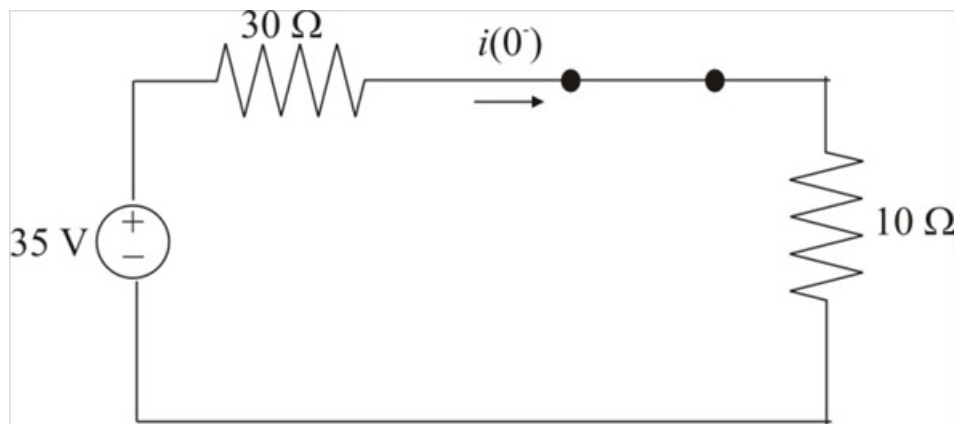


Figure 1

Step 2 of 8

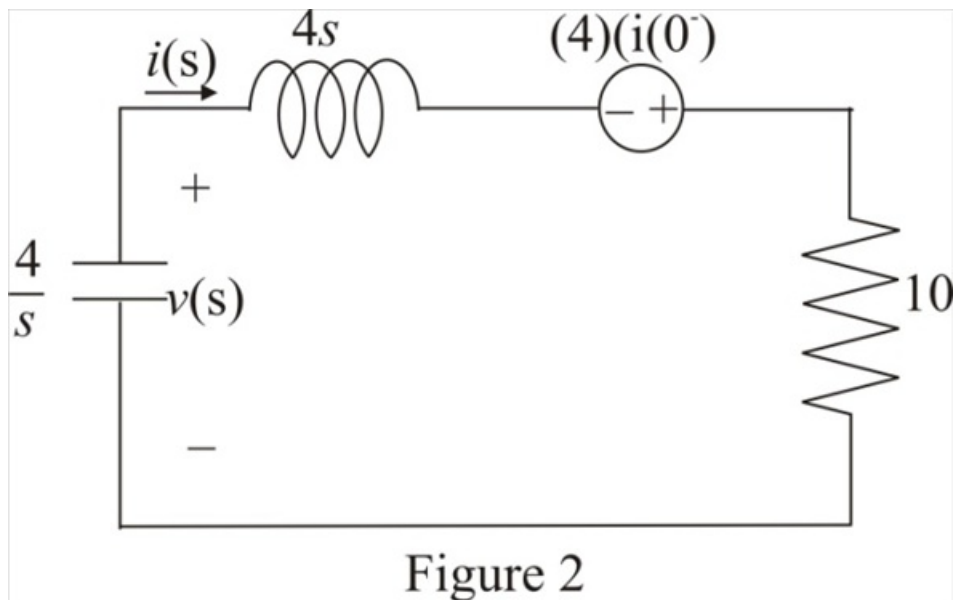
Calculate the value of the current $i(0^-)$.

$$\begin{aligned} i(0^-) &= \frac{35}{30+10} \\ &= \frac{35}{40} \\ &= 0.875 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(0^-)$ is 0.875 A.

Step 3 of 8

Draw the circuit diagram at $t > 0$. Represent the circuit parameters in Laplace transform.



Step 4 of 8

Apply Kirchhoff's Voltage Law in the loop.

$$i(s) \left(4s + \frac{4}{s} + 10 \right) - (4)(i(0^-)) = 0$$

Substitute 0.875 A for $i(0^-)$.

$$i(s) \left(4s + \frac{4}{s} + 10 \right) - (4)(0.875) = 0$$

$$i(s) \left(\frac{4s^2 + 10s + 4}{s} \right) - 3.5 = 0$$

$$i(s)(4s^2 + 10s + 4) = 3.5s$$

$$i(s) = \frac{3.5s}{4s^2 + 10s + 4}$$

$$i(s) = \frac{3.5s}{4(s^2 + 2.5s + 1)}$$

$$= \frac{0.875s}{s^2 + 2.5s + 1}$$

Therefore, the value of the current $i(s)$ is $\frac{0.875s}{s^2 + 2.5s + 1}$.

Step 5 of 8

Calculate the value of the voltage $v(s)$.

$$v(s) = -i(s) \left(\frac{4}{s} \right)$$

Substitute $\frac{0.875s}{s^2 + 2.5s + 1}$ for $i(s)$.

$$\begin{aligned} v(s) &= - \left(\frac{0.875s}{s^2 + 2.5s + 1} \right) \left(\frac{4}{s} \right) \\ &= \frac{-3.5}{s^2 + 2.5s + 1} \end{aligned}$$

$$v(s) = \frac{-3.5}{(s+0.5)(s+2)} \dots\dots (1)$$

Step 6 of 8

Assume $\frac{-3.5}{(s+0.5)(s+2)}$ to be $\frac{A_1}{s+0.5} + \frac{A_2}{s+2}$.

$$\frac{-3.5}{(s+0.5)(s+2)} = \frac{A_1}{s+0.5} + \frac{A_2}{s+2}$$

Apply residue method to find the value of the constant A_1 .

$$\begin{aligned} A_1 &= (s+0.5) \left(\frac{-3.5}{(s+0.5)(s+2)} \right) \Bigg|_{s=-0.5} \\ &= \frac{-3.5}{s+2} \Bigg|_{s=-0.5} \\ &= \frac{-3.5}{-0.5+2} \\ &= \frac{-3.5}{1.5} \end{aligned}$$

$$A_1 = -2.333$$

Therefore, the value of the constant A_1 is -2.333 .

Step 7 of 8

Apply residue method to find the value of the constant A_2 .

$$\begin{aligned} A_2 &= (s+2) \left(\frac{-3.5}{(s+0.5)(s+2)} \right) \Bigg|_{s=-2} \\ &= \frac{-3.5}{s+0.5} \Bigg|_{s=-2} \\ &= \frac{-3.5}{-2+0.5} \\ &= \frac{-3.5}{-1.5} \end{aligned}$$

$$A_2 = 2.333$$

Therefore, the value of the constant A_2 is 2.333 .

Step 8 of 8

Recall equation (1).

$$v(s) = \frac{-3.5}{(s+0.5)(s+2)}$$

Substitute $\frac{A_1}{s+0.5} + \frac{A_2}{s+2}$ for $\frac{-3.5}{(s+0.5)(s+2)}$.

$$v(s) = \frac{A_1}{s+0.5} + \frac{A_2}{s+2}$$

Substitute 2.333 for A_2 and -2.333 for A_1 .

$$v(s) = \frac{-2.333}{s+0.5} + \frac{2.333}{s+2}$$

Apply inverse Laplace transform.

$$\begin{aligned} v(t) &= -2.333e^{-0.5t} + 2.333e^{-2t} \\ &= 2.333(e^{-2t} - e^{-0.5t}) \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $\boxed{2.333(e^{-2t} - e^{-0.5t}) \text{ V}}$.